# Math F113X: Homework Set 10 — Solutions

**Section 2.1.6:** # 1–13
**Section 2.2.11:** # 1–6, 8, 10

## Section 2.1.6: Introduction to Spreadsheets

### Problem 1

Convert 4/7 to a decimal.

**Solution**

Type into a spreadsheet: `=4/7`

**Answer:** 0.5714285714

### Problem 2

Convert 16% to a decimal.

**Solution**

Type into a spreadsheet: `=16/100`

**Answer:** 0.16

### Problem 3

Add 8 and 19.

**Solution**

Type into a spreadsheet: `=8+19`

**Answer:** 27

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## Problem 4

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Find the difference of 230 and 78.

**Solution**

Type into a spreadsheet: `=230-78`

**Answer:** 152

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## Problem 5

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Multiply 12 and 9.

**Solution**

Type into a spreadsheet: `=12*9`

**Answer:** 108

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## Problem 6

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Divide 0.09 by 52.

**Solution**

Type into a spreadsheet: `=0.09/52`

**Answer:** 0.001730769231

## Problem 7

Calculate $8^3$.

**Solution**

Type into a spreadsheet: `=8^3`

**Answer:** 512

## Problem 8

Your bill at a restaurant is \$55.75 and you want to leave a 20% tip. How much would you add to the bill?

**Solution**

Type into a spreadsheet: `=55.75*0.2`

**Answer:** 11.15

## Problem 9

You leave a tip for \$7.50 for a bill at a restaurant that is \$44.50. What percent tip did you leave?

**Solution**

Type into a spreadsheet: `=7.50/44.50`

**Answer:** 0.1685393258

This is a 16.9% tip. I obtained this by the calculation: 7.50/44.50.

## Problem 10

In Column A use the fill down feature to build a spreadsheet starting with \$5 and ending at \$125, in increments of \$5. In Column B write a formula with a cell reference to calculate a 15.5% tip on the amount in Column A. Use the fill down feature to complete your table.

**Solution**

Type into a spreadsheet:

`A1` Bill
`B1` Tip Amount
`A2` `5`
`A3` `=A2+5` (fill down to A26)
`B2` `=A2*0.155` (fill down to B26)

| Bill | Tip Amount |
|---|---|
| \$5 | 0.775 |
| \$10 | 1.55 |
| \$15 | 2.325 |
| \$20 | 3.1 |
| \$25 | 3.875 |
| \$30 | 4.65 |
| \$35 | 5.425 |
| \$40 | 6.2 |
| \$45 | 6.975 |

| Bill | Tip Amount |
|---|---|
| \$50 | 7.75 |
| \$55 | 8.525 |
| \$60 | 9.3 |
| \$65 | 10.075 |
| \$70 | 10.85 |
| \$75 | 11.625 |
| \$80 | 12.4 |
| \$85 | 13.175 |
| \$90 | 13.95 |

| Bill | Tip Amount |
|---|---|
| \$95 | 14.725 |
| \$100 | 15.5 |
| \$105 | 16.275 |
| \$110 | 17.05 |
| \$115 | 17.825 |
| \$120 | 18.6 |
| \$125 | 19.375 |

## Problem 11

Imagine a certain savings account started out with a balance of \$5250.00 on day-one, and today has a current balance of \$5780.23.

(a) Exactly how much more money does the account have today, compared with day-one?

(b) Rounding to the nearest tenth of a percent: By what percentage amount has the account balance grown?

(c) If instead, the bank balance today was exactly double the starting balance, then by what exact percentage amount would the bank balance have grown?

(d) If the bank balance today had instead grown by 15.5% since day-one, then what would be the exact amount of today's balance?

**Solution**

### (a)

Type into a spreadsheet: `=5780.23-5250`

**Answer:** 530.23

### (b)

Type into a spreadsheet: `=530.23/5250`

**Answer:** 0.101, or 10.1%

### (c)

Type into a spreadsheet: `=5250/5250`

**Answer:** 100%

*Explanation:* The account would grow by 5250. Thus $5250/5250 = 1$, or 100%.

### (d)

Type into a spreadsheet: `=5250*(1.155)`

**Answer:** 6063.75

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## Problem 12

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Imagine that at the start of a certain month, you will make an opening deposit of $500 into a savings account, and you will then leave the account alone (meaning you will make no further deposits or withdrawals). Also, for this account: Every month after the opening deposit, the

amount in the account will grow to be 101% of its previous month's balance.

(a) Use a spreadsheet to enter 500 in cell A1. Using a formula and a cell-reference: Compute in cell A2, the amount in the account after one month has passed. Then using the fill down feature, continue the pattern for another eleven full months (you should end at cell A13). Format all the cells to show dollar signs. What is the amount in the account after one year?

(b) Now continue the pattern in column A of your spreadsheet to extend for a second full year (you should end at cell A25). What is the amount in the account after two years?

(c) What overall percentage growth occurred in the account between the opening deposit and one year later? (Compute using a formula and cell references)

(d) What overall percentage growth occurred in the account between the end of year one, and the end of year two? (Compute using a formula and cell references)

(e) (Challenge) The annual percentage growth that you found in part (d) for the second year, should be identical to the annual percentage growth that you found in part (c) for the first year. Can you mathematically explain why this is true? Do you think this pattern of identical overall annual percentage growth would continue, if you extend the pattern for even more years?

**Solution**

## (a)

Type into a spreadsheet:

```
A1   500
A2   =A1*1.01    (fill down to A13)
```

**Answer:** $563.41

| month | beginning of month | end of month |
|---|---|---|
| 0 | | 500.00 |
| 1 | 500.00 | 505.00 |
| 2 | 505.00 | 510.05 |
| 3 | 510.05 | 515.15 |
| 4 | 515.15 | 520.30 |
| 5 | 520.30 | 525.51 |
| 6 | 525.51 | 530.76 |
| 7 | 530.76 | 536.07 |
| 8 | 536.07 | 541.43 |
| 9 | 541.43 | 546.84 |
| 10 | 546.84 | 552.31 |
| 11 | 552.31 | 557.83 |
| 12 | 557.83 | 563.41 |

| month | beginning of month | end of month |
|---|---|---|
| 13 | 563.41 | 569.05 |
| 14 | 569.05 | 574.74 |
| 15 | 574.74 | 580.48 |
| 16 | 580.48 | 586.29 |
| 17 | 586.29 | 592.15 |
| 18 | 592.15 | 598.07 |
| 19 | 598.07 | 604.05 |
| 20 | 604.05 | 610.10 |
| 21 | 610.10 | 616.20 |
| 22 | 616.20 | 622.36 |
| 23 | 622.36 | 628.58 |
| 24 | 628.58 | 634.87 |

### (b)

Type into a spreadsheet:

`A14` fill down from A13 to A25

**Answer:** $634.87

### (c)

Type into a spreadsheet: `=(A13-A1)/A1`

**Answer:** 0.1268250301, or 12.68%

### (d)

Type into a spreadsheet: `=(A25-A13)/A13`

**Answer:** 0.1268250301

**(e)** There are two ways to think about this. One is strictly mathematical. The *percent* gained each month is $\dfrac{P(1.01) - P}{P} = 0.01$ no matter what $P$ is! An alternative way to say it is that even though we gained more money in year 2 compared to year 1 (roughly $71 compared to $63), we started with more money in year 2. So you can gain more money overall but the percent gain can be the same.

Yes. Every year the balance is multiplied by $1.01^{12}$, whatever it started at, so every year shows the same 12.68% growth.

## Problem 13

Imagine that at the start of a certain year, you will deposit $1000.00 into a savings account, and then you will leave the account alone. Each year after the opening deposit, the amount in the account will grow to be 103% of its previous year's balance.

(a) After two years, the account balance will have experienced two growth amounts of 103%. You can find this account balance amount here, with the spreadsheet computation `= 1000 * (103%) * (103%)`. Perform this computation in a spreadsheet and write the balance that you find.

(b) Now enter the spreadsheet computation `= 1000 * (103%)^2`. Notice that the result here, which involves using a power, gives the same answer as you found in part (a). Comparing the two spreadsheet computations: Explain why they give the same result.

(c) Using the pattern in part (b) above, and carefully choosing the power: Compute the balance that will be in the account fifteen full years after the account was originally opened. (Round to the nearest cent)

(d) (Challenge) Make a spreadsheet that shows the account balance each individual year for 30 years. From the date of the opening deposit: What minimum number of full years will you have to wait, until the balance finally exceeds twice its opening deposit amount? (Use cell references, the fill down feature, and dollar formatting)

(e) (Challenge) Imagine the opening balance of the account was $5000.00 instead of $1000.00 (and everything else about the account stays the same). Make a similar spreadsheet as you did in part (d), and using this spreadsheet, find the minimum number of full years you will have to wait this time, until the balance finally exceeds twice its opening deposit amount. How does this answer compare with your answer in part (d)? Do you think your answer would be the same here, for any positive opening balance you may choose for the account?

**Solution**

### (a)

Type into a spreadsheet: `=1000*(1.03)*(1.03)`

**Answer:** $1,060.90

### (b)

Type into a spreadsheet: `=1000*(1.03)^2`

**Answer:** $1,060.90

Raising to the power 2 means multiplying by the number twice, so $1000 \cdot (1.03)^2$ and $1000 \cdot (1.03) \cdot (1.03)$ are the same computation.

### (c)

Type into a spreadsheet: `=1000*(1.03)^15`

**Answer:** $1,557.97

## (d)

Type into a spreadsheet:

`A1` `0`
`B1` `1000`
`A2` `=A1+1` (fill down to A31)
`B2` `=B1*1.03` (fill down to B31)

**Answer:** 24 years

| year | end of year balance |
|---|---|
| 0 | $1,000.00 |
| 1 | $1,030.00 |
| 2 | $1,060.90 |
| 3 | $1,092.73 |
| 4 | $1,125.51 |
| 5 | $1,159.27 |
| 6 | $1,194.05 |
| 7 | $1,229.87 |
| 8 | $1,266.77 |
| 9 | $1,304.77 |
| 10 | $1,343.92 |

| year | end of year balance |
|---|---|
| 11 | $1,384.23 |
| 12 | $1,425.76 |
| 13 | $1,468.53 |
| 14 | $1,512.59 |
| 15 | $1,557.97 |
| 16 | $1,604.71 |
| 17 | $1,652.85 |
| 18 | $1,702.43 |
| 19 | $1,753.51 |
| 20 | $1,806.11 |
| 21 | $1,860.29 |

| year | end of year balance |
|---|---|
| 22 | $1,916.10 |
| 23 | $1,973.59 |
| 24 | $2,032.79 |
| 25 | $2,093.78 |
| 26 | $2,156.59 |
| 27 | $2,221.29 |
| 28 | $2,287.93 |
| 29 | $2,356.57 |
| 30 | $2,427.26 |

## (e)

Type into a spreadsheet:

`A1` `0`
`B1` `5000`
`A2` `=A1+1` (fill down to A31)
`B2` `=B1*1.03` (fill down to B31)

**Answer:** 24 years

*Calculation:* I'm going to check by a quick guess.

Type into a spreadsheet: `=5000*(1.03)^24` output: $10,163.97

This is the same as the answer in part (d), and it would be the same for any positive opening balance: the balance after $n$ years is (opening balance)$\times 1.03^n$, so it passes double the opening balance exactly when $1.03^n$ passes 2, which first happens at $n = 24$.

## Section 2.2.11: Simple and Compound Interest

## Problem 1

A friend lends you $200 for a week, which you agree to repay with 5% one-time interest. How much will you have to repay?

**Solution**

Type into a spreadsheet: `=200*(1+0.05)`

**Answer:** $210.00

## Problem 2

You loan your friend $100. They agree to pay an annual interest rate of 3%, simple interest. Six months later they repay that loan.

(a) How much did they pay you?

(b) How much was interest?

**Solution**

### (a)

Type into a spreadsheet: `=100*(1+0.03/2)`

**Answer:** $101.50

### (b)

Type into a spreadsheet: `=B11-100` (B11 holds the answer to (a))

**Answer:** $1.50

## Problem 3

Consider a simple interest loan of \$200 with an annual interest rate of 6%. If that loan is paid off 1 year and 3 months later, how much was repaid?

**Solution**

Type into a spreadsheet: `=200*(1+0.06*(1.25))`

**Answer:** \$215.00

The reasoning here is that 1 year and 3 months is 1.25 of a year.

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## Problem 4

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You deposit \$1,000 in an account that earns simple interest. The annual interest rate is 2.5%.

(a) How much interest will you earn in 5 years?

(b) How much will you have in the account in 5 years?

**Solution**

### (a)

Type into a spreadsheet: `=1000*(0.025*5)`

**Answer:** \$125.00

### (b)

Type into a spreadsheet: `=1000*(1+0.025*5)`

**Answer:** \$1,125.00

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## Problem 5

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Consider an investment of \$20000 with an annual interest rate of 5%.

(a) If that investment is earning simple interest, how much will the investment be worth in 10 years?

(b) If that investment is getting annually compounding interest, how much will the investment be worth in 10 years?

**Solution**

**(a)**

Type into a spreadsheet: `=20000*(1+0.05*10)`

**Answer:** $30,000.00

**(b)**

Type into a spreadsheet: `=20000*(1+0.05)^10`

**Answer:** $32,577.89

## Problem 6

Nico invests $4,500 into an account that has an annual interest rate of 8.5%. The interest is compounding monthly. Twenty years later what is the account balance?

**Solution**

Type into a spreadsheet: `=4500*(1+0.085/12)^(12*20)`

**Answer:** $24,485.59

## Problem 8

Suppose you obtain a $3,000 Certificate of Deposit (CD) with a 3% APR, paid quarterly, with maturity in 5 years.

(a) What is the future value of the CD in 5 years?

(b) How much interest will you earn?

(c) What percent of the balance is interest?

Solution

**(a)**

Type into a spreadsheet: `=3000*(1+0.03/4)^(4*5)`

**Answer:** $3,483.55

**(b)**

Type into a spreadsheet: `=B45-3000` (B45 holds the answer to (a))

**Answer:** $483.55

**(c)**

Type into a spreadsheet: `=B48/B45` (B48 and B45 hold the answers to (b) and (a))

**Answer:** 13.88%

or type `=483.55/3483.55`

Observe that 13.88% is a lot more than 3%.

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## Problem 10

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You deposit $2,000 in an account earning 3% APR compounded monthly.

(a) How much will you have in the account in 20 years?

(b) How much interest will you earn?

(c) What percent of the balance is interest?

(d) What percent of the balance is the principal?

Solution

**(a)**

Type into a spreadsheet: `=2000*(1+0.03/12)^(12*20)`

**Answer:** $3,641.51

**(b)**

Type into a spreadsheet: `=B57-2000` (B57 holds the answer to (a))

**Answer:** $1,641.51

### (c)

Type into a spreadsheet: `=B60/B57` (B60 and B57 hold the answers to (b) and (a))

**Answer:** 45.08%

That's a lot!

### (d)

Type into a spreadsheet: `=2000/B57` (B57 holds the answer to (a))

**Answer:** 54.92%